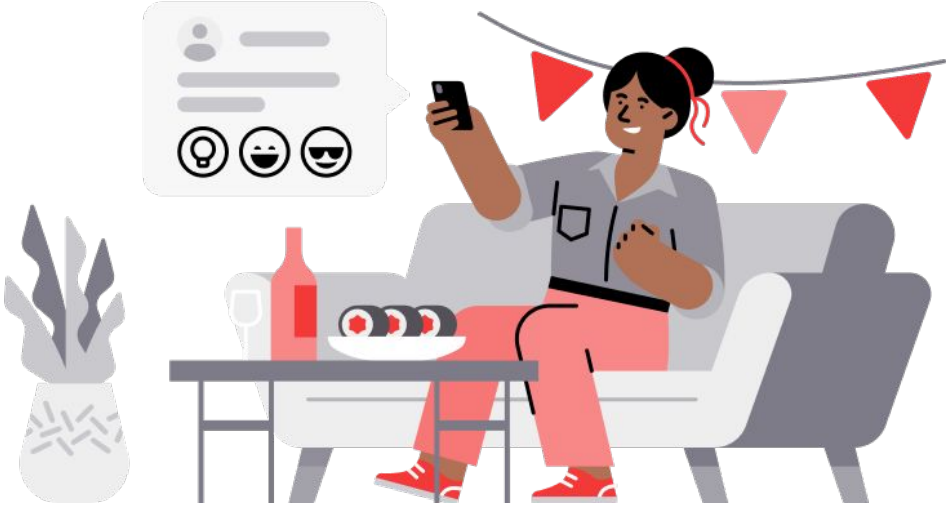
Consumer | Contributions

Recognition badges

2/07/2023



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Overview

Overview

Background

- Recognition in form of feedback pushes have shown increased reviewer retention and reviews
- Badges can be another form of recognition that is more tangible and we wanted to test if recognizing users with badges in topic of interest can also motivate users to contribute more

Hypothesis

- Telling users that they are close to earning a badge in the category of interest (and rewarding them badges) will motivate them to write another review in that category

Goal for this experiment

- Increase reviews from existing reviewers

What we tested

1. **Determine which badges to create:**
   - Top ~50 RFN sub-categories with highest # of repeat reviewers who wrote min 2 reviews of same sub category in the last year (~500K users)

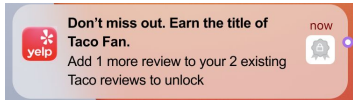
Updated Name	With "Fan"	With a preceding article
Brunch	Brunch Fan	a Brunch Fan
Seafood	Seafood Fan	a Seafood Fan
Sandwich	Sandwich Fan	a Sandwich Fan
Pizza	Pizza Fan	a Pizza Fan
Coffee	Coffee Fan	a Coffee Fan
Burger	Burger Fan	a Burger Fan
Cocktail	Cocktail Fan	a Cocktail Fan
Hot Dog	Hot Dog Fan	a Hot Dog Fan
Sushi	Sushi Fan	a Sushi Fan
Bakery	Bakery Fan	a Bakery Fan
Steak	Steak Fan	a Steak Fan
BBQ	BBQ Fan	a BBQ Fan
Salad	Salad Fan	a Salad Fan
Cafe	Cafe Fan	a Cafe Fan
Chicken Wings	Chicken Wings F	a Chicken Wings Fan
Ice Cream	Ice Cream Fan	an Ice Cream Fan
Wine Bar	Wine Bar Fan	a Wine Bar Fan
Juice Bar	Juice Bar Fan	a Juice Bar Fan
Noodles	Noodles Fan	a Noodles Fan
Bubble Tea	Bubble Tea Fan	a Bubble Tea Fan
Sports Bar	Sports Bar Fan	a Sports Bar Fan
Ramen	Ramen Fan	a Ramen Fan
Taco	Taco Fan	a Taco Fan
Delis	Delis Fan	a Delis Fan
Soup	Soup Fan	a Soup Fan
Diner	Diner Fan	a Diner Fan
Brewery	Brewery Fan	a Brewery Fan
Donut	Donut Fan	a Donut Fan
Food Truck	Food Truck Fan	a Food Truck Fan
Cajun	Cajun Fan	a Cajun Fan
Tapas	Tapas Fan	a Tapas Fan
Comfort Food	Comfort Food Fa	a Comfort Food Fan
Halal	Halal Fan	a Halal Fan
Dim Sum	Dim Sum Fan	a Dim Sum Fan

What we tested

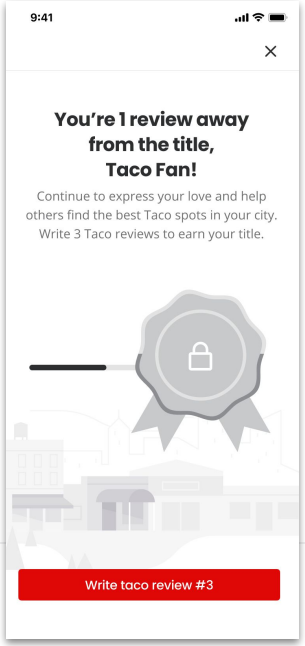
2. **Let reviewer know an additional review will get them recognized as a expert/specialist in the topic.**
   There were 2 user flows in which we let users know about earning a badge:

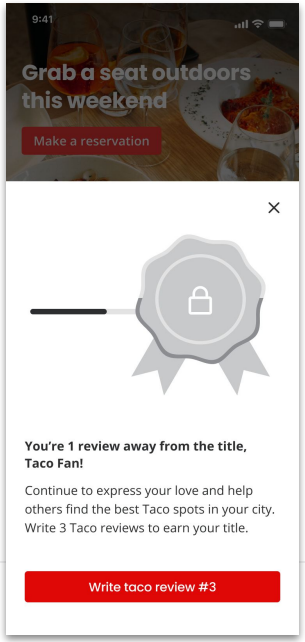
A) Push/modal for users who already wrote 2 reviews for same biz category

Cohort 1



Cohort 2





Modal was shown only only for users who have push disabled

What we tested

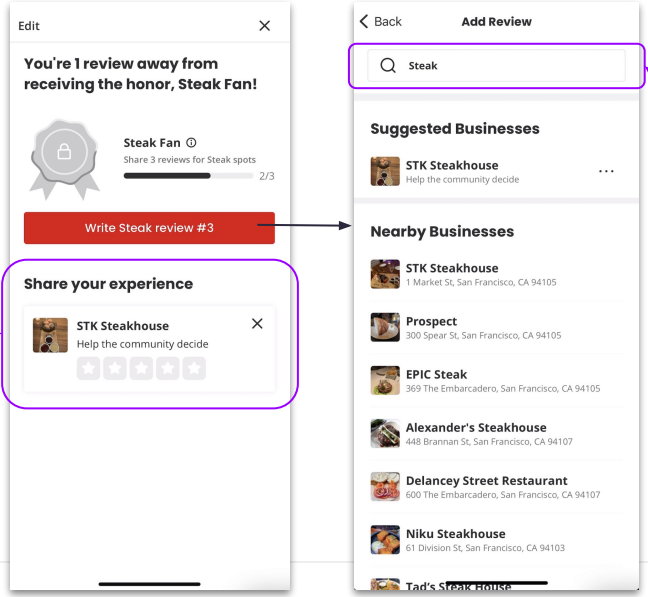
2. **Let reviewer know an additional review will get them recognized as a expert/specialist in the topic.**
   There were 2 user flows in which we let users know about earning a badge:

B) Post-review page for users who just completed their 2nd review for the same category

If user has YNRA suggestions in the same badge category, then only show those.

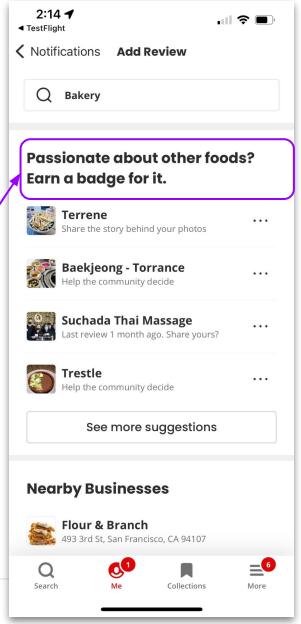
If not, then update "share your experience" header of YNRA section to something more encouraging of starting another badge (see below)

Passionate about other foods? Earn a badge for it.



WAR flow was also pre-filtered to the same category to only show businesses in that category under "nearby businesses"

Just like on the post-review page, we update the title for YNRA if user does not have same category YNRA suggestions

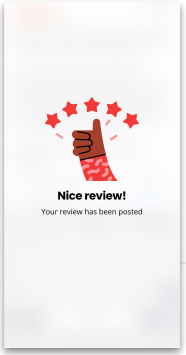


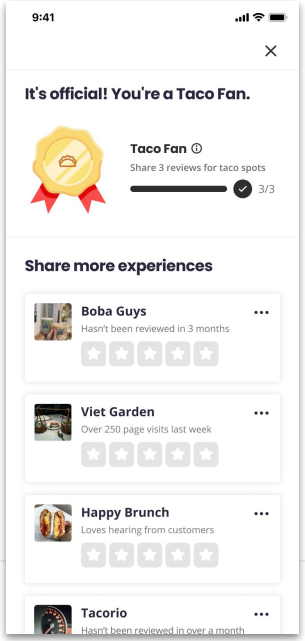
What we tested

3. **Reward reviewer with expert recognition badge.** We had both immediate and revisitable messages for recognition of badges:

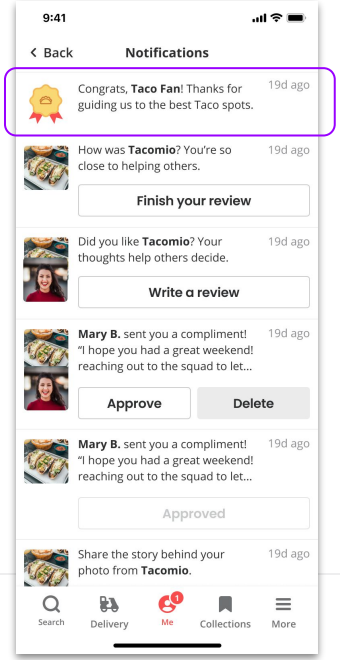
A) Post review (immediate) recognition
We updated the success screen for reviews posted

Status Quo - Success screen

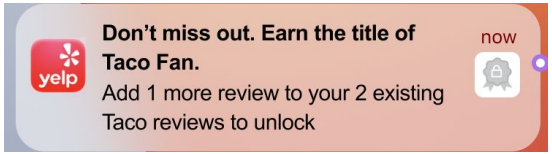




B) In-app notification (revisitable) recognition
Users can revisit badge recognition message from notification page



Experiment setup

Status quo Users who already wrote 2 reviews for badge eligible biz categories, and users who just wrote their 2nd review for badge eligible biz categories but do not see push/modal nor post review page about writing 1 more to earn a badge	**Cohort 1: more formal copy ("Earn the title...")** Users who already wrote 2 reviews for badge eligible biz categories, *and* users who just wrote their 2nd review for badge eligible biz categories And see push/modal or the post review page about writing 1 more to earn a badge	**Cohort 2: More playful copy ("We see you...")** Same as cohort 2 but different copy
		

Who was included in the test

Who	Repeat reviewers
Cohort logic	• Users who already wrote 2 reviews in past 12 months in badge eligible categories (one of the 50 rfn categories we made badges for) • Users who just wrote their 2nd review in badge eligible categories
Platform	iOS
Categories	RFN
When	11/16/2022 - 1/10/2023
Where	US only
Type of test	Simple AB

Population size:
244,897 users

Measurement strategy

scorecard

Primary	**# of reviews posted per user per day**	**Significant increase over SQ would validate hypothesis**
Guardrails		No harm

Results

TL;DR

- **The experiment validated the hypothesis that recognition badges do motivate existing reviewers to write their next review:**
  - Both treatment cohorts saw a lift in reviews written compared to SQ:
    - +5.54% (estimated **+20.5K reviews/mo**) and +7.83% (estimated **+29K reviews/mo**)
- **Engagement was really high among users who received a message about getting a badge**
  - 20% of users who got badge message actually earned a badge (47K users) with about half of them earning more than 1 badge (23K users)
  - Total of about **69K badges were awarded**
- **Reviews written for badges were of higher quality than regular reviews (not for badges)**
  - **+6-8% higher Yelp Sort score** for reviews written for badges than not
  - Over 80% of reviews written for badges were recommended

*** Indicate statistically significant results with an asterisk**

Metrics report

[scorecard]

		Cohort 1: "***Don't miss out. Earn the title of Taco Fan.*** *Add 1 more review to your 2 existing Taco reviews to unlock"*		Cohort 2: "***We see you, Taco Fan.*** *Write 1 more review to receive the honor of being called a Taco Fan"*	
	Status quo	**Lift relative to SQ**	**Lift per month @ 100%**	**Lift relative to SQ**	**Lift per month @ 100%**
Reviews posted per day per user avg	**0.026**	**+5.54%**	**+20.5K reviews** [14.1K, 27.5K] 95% CI	**+7.83%**	**+29K reviews** [22K, 36.7K] 95% CI *This is about 3% global lift (RFN reviews)*

Analysis follow ups

Higher engagement than other review channels and likely no novelty effect

- **~20% (47K* users) of users** who received badge message **earned a badge** (out of 238.6K users)
  - Avg feedback push CTR 8%, review solicitation push CTR 2% (both have much lower % for users who actually wrote reviews)
- **~50% (23K* users)** of users who earned a badge **earned more than 1 badge**
  - Indicative of this not being novelty effect (1-time thing) but intentional and continued motivation
- Total of **69K* badges were awarded**

*actual # of badges earned during experiment (60 days)

What type of badges were most awarded?

More unique, specific dish-type subcategories were awarded the least (likely due to categorization of businesses on Yelp)

Brunch	8288 badges
Seafood	5784
Pizza	4818
Coffee	4799
Sandwich*	4472
Cocktail	3699
Burger	3448
Sushi	2839
Bakery	2245
Fast Food	2145
Steak	1997
BBQ	1744
Salad	1510
Cafe	1289
Noodles	1192
Ice Cream	1100
Wine Bar	1080
Tea	1072
Bubble Tea	1029

Chicken Wings	858
Ramen	820
Juice Bar	755
Brewery	741
Sports Bar	732
Taco	417
Donut	383
Soup	371
Diner	360
Dim Sum	357
Cajun	321
Hot Pot	300
Food Truck	268
Tapas	268
Halal Food	224
Tex-Mex	191
Bagel	152
Comfort Food	151

Poke	106
Acai Bowl	82
Soul Food	60
Shaved Ice	55
Izakaya	52
Cake Shop	44
Cupcake	38
Fish and Chips	21

Post review page was much more effective than push in getting users to write their badge review

Less than 1% of badge reviews came from push.

Review source	% of all reviews that led to badges
YNRAs on Post review page after 2nd review We filtered YNRAs to show only bizzes of same badge category	12.2%
“Write review #3” button on Post review page after 2nd review	3.4%
YNRAs on Post review page after 3rd review This is after user earned a badge, they wrote another review from YNRA in the post review page	1.8%
“Write review #3” button on push landing page	0.2%

Q: should we not do pushes for badges?
Something to consider; badges may be more of in-product feature/experience

Yelp sort score is higher for reviews written by users who got badges than for reviews written by users in SQ*

- **Lift relative to SQ (score: 3.9):**
  - cohort 1: 4.3 score, +8.1% [7.5, 8.6]% 95% CI
  - cohort 2: 4.2 score, +6.2% [5.6, 6.7]% 95% CI

- **81% of reviews written for badges were recommended**
  - Avg recommendation rate is 80% for reviews

*includes only 3rd reviews in RFN sub-categories that are badge eligible in the treatment cohorts

Next steps

1. ***We are not shipping to 100%*** because flow is not end-to-end, *but* we want to move forward with building badges on Yelp

2. **Badge framework:** (presenting next Tue)
   a. The flywheel and network effects of badges (what value do badges bring to our contributors, readers, Yelp)
   b. Levers that make up the badge ecosystem on Yelp: quantity, quality, continuous motivation, ornamentation of badges
   c. V1 MVP

Appendix

Takeaway

- Knowing the close potential to earn a badge does motivate existing reviewers to write another review in the category/topic of their interest
- Badge messaging in integrated flows like post review page can be more user friendly and effective for users to take the next step in earning a badge
- Users wrote higher quality reviews on average for badges than without badges (badge can not only motivate user to contribute more but to contribute with more thought/care since it’s topic of their interest)
- Categorization of businesses on Yelp make it hard to reward badges for very specific categories

+26K/mon* more reviews written in treatment than SQ

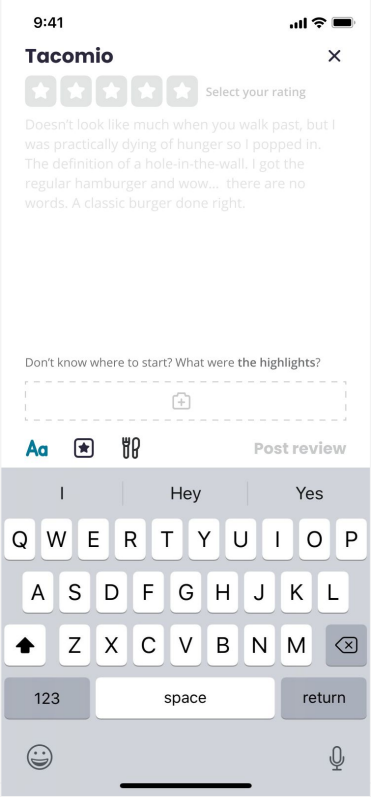
- In 30 days at 100%, there were about 58K reviews written that got users badges, but the additional # of reviews in treatment than in SQ is about +26K reviews.
- We speculate that the 32K reviews (58K-26K) are from users who are already motivated to write more reviews regardless of badges in SQ, and the +26K reviews were from more casual/infrequent reviewers who were motivated to write more reviews because of badges.

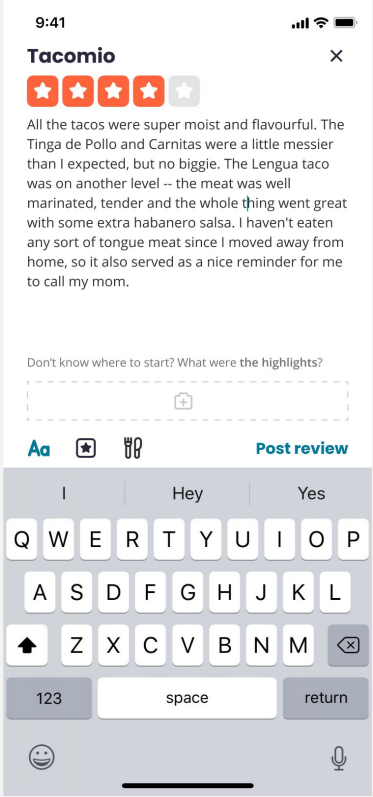
* normalized at 100% in 30 days for fair comparison between treatment and SQ

Engagement

	CTR	Review through rate	note
Cohort 1 push	Avg 8.73%		
Cohort 2 push	Avg 9.05%		
Post review “Write review 3” CTA	6.8%		
In-app msg about writing another review to get a badge	1.05%		Low CTR is not surprising - the pushes and post review screen message about writing another review was also sent as in-app push so users would have already engaged with the push or post review page first

Status Quo


9:41
Tacomio
Select your rating
Doesn't look like much when you walk past, but I was practically dying of hunger so I popped in. The definition of a hole-in-the-wall. I got the regular hamburger and wow... there are no words. A classic burger done right.
Don't know where to start? What were the highlights?
Aa
Post review
I
Hey
Yes
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
space
return


9:41
Tacomio
All the tacos were super moist and flavourful. The Tinga de Pollo and Carnitas were a little messier than I expected, but no biggie. The Lengua taco was on another level -- the meat was well marinated, tender and the whole thing went great with some extra habanero salsa. I haven't eaten any sort of tongue meat since I moved away from home, so it also served as a nice reminder for me to call my mom.
Don't know where to start? What were the highlights?
Aa
Post review
I
Hey
Yes
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
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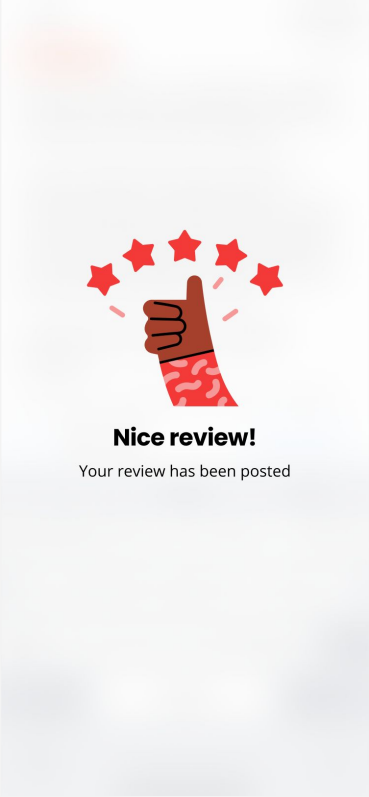
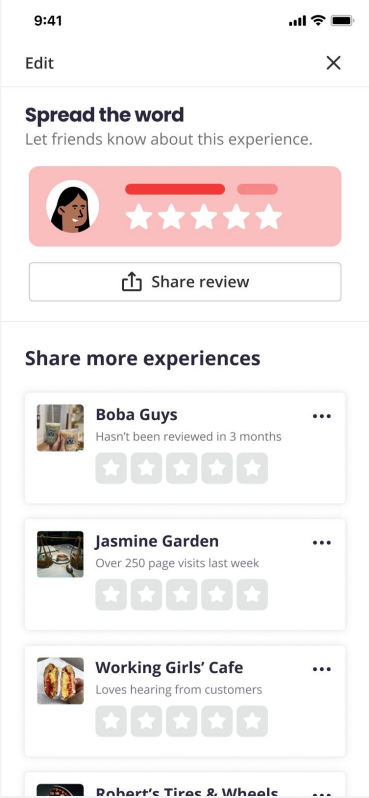

Nice review!
Your review has been posted


9:41
Edit
Spread the word
Let friends know about this experience.
Share review
Share more experiences
Boba Guys
Hasn't been reviewed in 3 months
Jasmine Garden
Over 250 page visits last week
Working Girls' Cafe
Loves hearing from customers

Links

Experiment resources

PEP

Product Page Wiki

Of sample of 500 reviews that awarded badges, only X% were below standard review quality

Integrated flow like the post-review-page was much more effective than push in getting users to write their badge review

- **87x more badge reviews came from post review screen** than push landing page (both had "write review #3" CTAs)
- Most reviews came from **biz page review entry points** (perhaps more users were motivated in general after seeing badge message via push and post review pae)

Review source	% of all reviews that led to badges
Biz page question-based review entry point	18%
Biz page review CTA from contribution cell	15.4%
YNRAs on Post review page after 2nd review We filtered YNRAs to show only bizzes of same badge category	12.2%
Profile YNRA	9.8%
"Write review #3" button on Post review page after 2nd review	3.4%
YNRAs on Post review page after 3rd review This is after user earned a badge, they wrote another review from YNRA in the post review page	1.8%
"Write review #3" button on push landing page	0.2%